

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1153

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing techniques. (BL4,BL5)
Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:40

Annexure A

MOCK INTERVIEW

Code of Conduct:

I D RAGHU VARDHAN 192311051___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D RAGHU VARDHAN

RUBRICS FOR CALCULATION:

MOCK INTERVIEW

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

Q1. Scenario: Online Learning Platform

Modules:

User Login, Course Access, Quiz Module

Question:

Explain how **quality assurance techniques** can be applied to ensure reliable course access and accurate quiz evaluation. Provide a practical example.

Q1. Quality Assurance Techniques for an Online Learning Platform

Introduction

An **Online Learning Platform** consists of important modules such as **User Login, Course Access, and Quiz Module**. To provide a reliable learning experience, the system must ensure that students can access courses without errors and receive accurate quiz results. **Quality Assurance (QA)** is a systematic process used to verify that software meets specified requirements and functions correctly. QA techniques help identify defects early, improve software quality, and ensure system reliability.

Quality Assurance Techniques for Reliable Course Access

The **Course Access Module** allows students to view course materials, videos, assignments, and learning resources. QA techniques are applied to ensure that students can access content smoothly and without interruption.

1. Functional Testing

Functional testing verifies whether all course access features work according to requirements.

Testing Activities:

- Verify student login before course access.
- Check whether enrolled students can access courses.
- Verify course content loading correctly.
- Ensure unauthorized users cannot access restricted courses.

Possible Defects:

- Course page not opening.
- Incorrect course displayed.
- Unauthorized access granted.

Benefits:

- Ensures correct system functionality.
 - Improves user satisfaction.
-

2. Integration Testing

Integration testing verifies interaction between modules such as Login and Course Access.

Testing Activities:

- Check whether successful login redirects students to enrolled courses.
- Verify communication between course database and application.

Benefits:

- Detects interface-related defects.
 - Ensures smooth data flow.
-

3. Performance Testing

Performance testing evaluates system responsiveness under different workloads.

Testing Activities:

- Measure course loading time.
- Test multiple users accessing courses simultaneously.
- Verify video streaming performance.

Benefits:

- Prevents slow response times.
 - Ensures system stability under heavy usage.
-

4. Security Testing

Security testing protects learning resources from unauthorized access.

Testing Activities:

- Verify authentication mechanisms.
- Check access permissions.
- Test session management.

Benefits:

- Protects course content.
 - Prevents unauthorized access.
-

Quality Assurance Techniques for Accurate Quiz Evaluation

The **Quiz Module** is responsible for conducting assessments and calculating scores accurately.

1. Functional Testing

Verifies whether quiz operations perform correctly.

Testing Activities:

- Check question display.
- Verify answer submission.
- Validate score calculation.

Possible Defects:

- Incorrect marks calculation.
- Missing questions.
- Submission failures.

Benefits:

- Ensures assessment accuracy.
 - Improves reliability of results.
-

2. Boundary Value Testing

Tests system behavior using minimum and maximum input values.

Example:

- Minimum marks = 0
- Maximum marks = 100

Benefits:

- Identifies calculation errors.
 - Improves system robustness.
-

3. Database Testing

Ensures quiz responses and scores are stored correctly.

Testing Activities:

- Verify answer storage.
- Check result retrieval.
- Validate score records.

Benefits:

- Prevents data loss.
 - Maintains result accuracy.
-

4. Regression Testing

Performed after system updates.

Testing Activities:

- Verify previous quiz functions still work.
- Ensure score calculation remains accurate after modifications.

Benefits:

- Prevents introduction of new defects.
 - Maintains software quality.
-

Practical Example

Consider an online course named "**Programming Fundamentals**".

Scenario

A student named Rahul logs into the platform and accesses the course.

QA Validation Steps

Step 1: Login Testing

- Enter valid username and password.
- Verify successful login.

Expected Result: Dashboard opens successfully.

Step 2: Course Access Testing

- Click "Programming Fundamentals."
- Verify course videos and study materials load correctly.

Expected Result: All course content is accessible without errors.

Step 3: Quiz Testing

- Start a quiz containing 10 questions.
- Answer 8 questions correctly.

Expected Result: System calculates score as:

Score = $8 \times 10 = 80$ Marks

Step 4: Result Verification

- Compare displayed marks with expected marks.

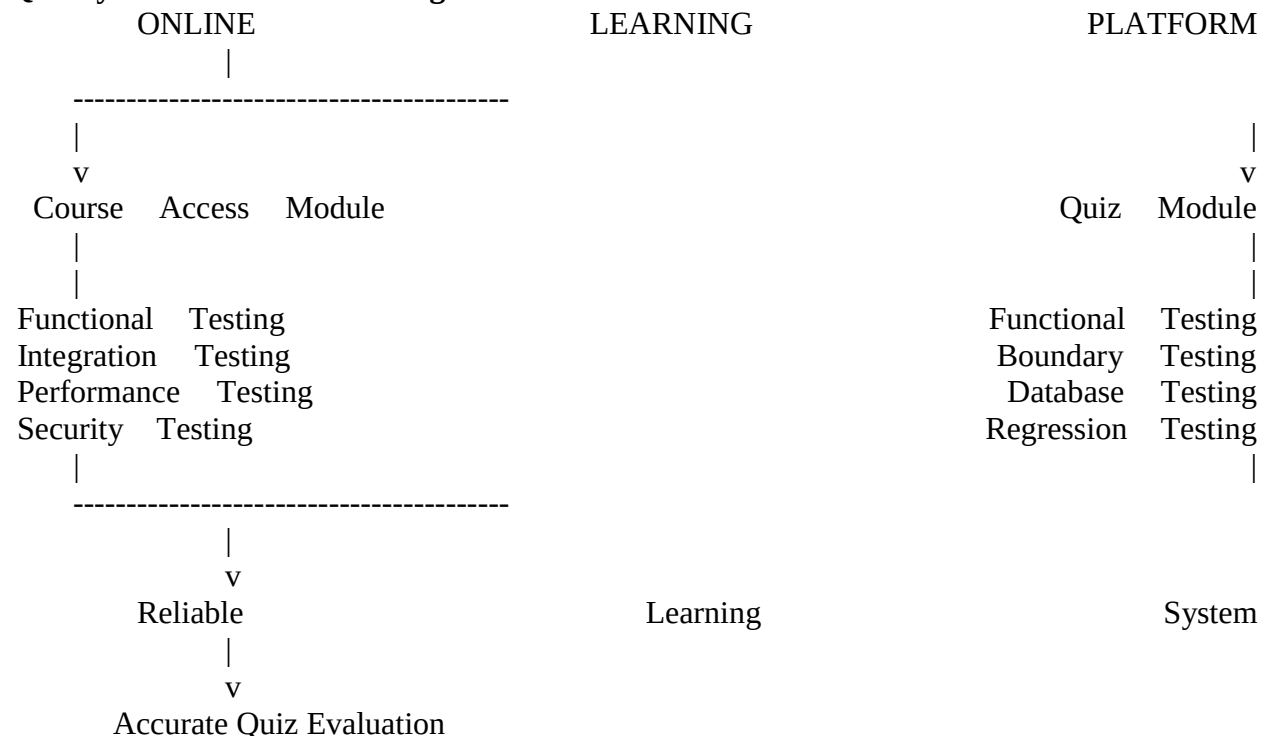
Expected Result: Final score displayed = 80.

Step 5: Database Verification

- Check whether score 80 is stored correctly in the database.

Expected Result: Database record matches displayed result.

Quality Assurance Process Diagram



Conclusion

Quality Assurance techniques play a vital role in ensuring the reliability of an Online Learning Platform. **Functional, Integration, Performance, and Security Testing** help guarantee smooth and secure course access, while **Functional, Boundary Value, Database, and Regression Testing** ensure accurate quiz evaluation. By applying these techniques, the platform delivers dependable course access, correct score calculation, improved user satisfaction, and high software quality.

(Suitable for 10–15 Marks University Examination Answer).

SOME WHAT BIG

Q1. Explain How Quality Assurance Techniques Can Be Applied to Ensure Reliable Course Access and Accurate Quiz Evaluation in an Online Learning Platform. Provide a Practical Example.

Introduction

An **Online Learning Platform** is a software system that enables students to access courses, watch learning materials, attempt quizzes, and track their progress online. The major modules of the system are **User Login, Course Access, and Quiz Module**. Since thousands of students may use the platform simultaneously, it is important to ensure that the system functions correctly and reliably. This is achieved through **Quality Assurance (QA)** techniques.

Quality Assurance is a systematic process of verifying and validating software to ensure that it meets user requirements, performs accurately, and remains free from defects. QA helps improve software reliability, security, performance, and user satisfaction.

Quality Assurance for Reliable Course Access

The **Course Access Module** allows students to access videos, notes, assignments, and learning resources. QA techniques are applied to ensure that course materials are available whenever students need them.

1. Functional Testing

Functional testing verifies whether all course access features work according to the specified requirements.

Activities Performed

- Verify successful login before course access.
- Check whether enrolled students can access their courses.
- Verify that course videos, PDFs, and assignments open correctly.
- Ensure unauthorized users cannot access premium or restricted courses.
- Verify navigation between different course sections.

Possible Defects

- Course page fails to load.
- Incorrect course content is displayed.
- Students are unable to download learning materials.
- Broken links within course modules.

Benefits

- Ensures proper functionality.
- Improves user experience.
- Reduces system failures.

2. Integration Testing

Integration testing verifies communication between different modules of the platform.

Activities Performed

- Verify interaction between Login Module and Course Access Module.
- Ensure user enrollment information is correctly retrieved from the database.
- Check communication between video servers and course pages.

Example

After login, the system should automatically display only the courses in which the student is enrolled.

Benefits

- Detects communication errors between modules.
- Ensures smooth workflow.

3. Performance Testing

Performance testing evaluates system behavior under different workloads.

Activities Performed

- Measure course page loading speed.
- Test video streaming quality.
- Simulate hundreds or thousands of students accessing courses simultaneously.
- Monitor server response time.

Possible Defects

- Slow page loading.
- Video buffering.
- Server crashes during peak usage.

Benefits

- Improves responsiveness.
 - Ensures platform stability.
-

4. Security Testing

Security testing protects educational content and student information.

Activities Performed

- Verify user authentication mechanisms.
- Check access control permissions.
- Test session timeout functionality.
- Detect unauthorized access attempts.

Benefits

- Prevents data breaches.
 - Protects learning resources.
 - Ensures secure user access.
-

Quality Assurance for Accurate Quiz Evaluation

The **Quiz Module** evaluates student knowledge and generates scores. Since quiz results affect student performance assessment, accuracy is critical.

1. Functional Testing

Functional testing verifies that quiz features operate correctly.

Activities Performed

- Verify question display.
- Check answer selection functionality.
- Validate quiz submission process.
- Verify score calculation.

Possible Defects

- Missing questions.
- Incorrect answer validation.
- Submission failures.
- Wrong score calculation.

Benefits

- Ensures accurate assessments.
 - Improves reliability.
-

2. Boundary Value Testing

Boundary Value Testing checks the system using minimum and maximum input values.

Example

If a quiz contains 20 questions:

- Minimum score = 0
- Maximum score = 100

The system should correctly calculate scores at both extremes.

Benefits

- Identifies calculation errors.
- Improves robustness.

3. Database Testing

Database testing ensures quiz responses and scores are stored correctly.

Activities Performed

- Verify answer storage.
- Check score recording.
- Validate result retrieval.

Possible Defects

- Missing records.
- Incorrect marks stored.
- Data corruption.

Benefits

- Ensures result integrity.
 - Prevents data loss.
-

4. Regression Testing

Regression testing is performed whenever new features or updates are introduced.

Activities Performed

- Verify previous quiz functionalities remain unaffected.
- Recheck score calculations after updates.
- Validate existing reports and results.

Benefits

- Prevents introduction of new defects.
 - Maintains software quality.
-

5. User Acceptance Testing (UAT)

User Acceptance Testing verifies that the system meets real user expectations.

Activities Performed

- Students attempt quizzes.
- Instructors review results.
- Feedback is collected.

Benefits

- Ensures practical usability.
 - Increases user satisfaction.
-

Practical Example

Consider an online course called **Python Programming Fundamentals**.

A student named Rahul is enrolled in the course and wants to access learning materials and attempt a quiz.

Step 1: User Login Testing

Rahul enters his username and password.

Expected Result

- Login is successful.
- Student dashboard opens.

QA Verification

- Validate credentials.
 - Check session creation.
 - Verify security controls.
-

Step 2: Course Access Testing

Rahul selects the Python Programming course.

Expected Result

- Course videos load correctly.

- Notes and assignments are available.
- Navigation works properly.

QA Verification

- Verify content availability.
- Measure loading time.
- Check video playback quality.

Step 3: Quiz Attempt Testing

Rahul starts a quiz containing 10 questions.

Suppose each question carries 10 marks.

Rahul answers 8 questions correctly.

Expected Calculation

Score = 8 × 10
= 80 Marks

Expected Result

The system should display **80 marks**.

QA Verification

- Check answer evaluation logic.
- Verify score calculation accuracy.
- Validate submission process.

Step 4: Result Storage Testing

After quiz submission, the score should be saved in the database.

Expected Result

Student Name : Rahul
Course : Python Programming
Score : 80 Marks
Status : Pass

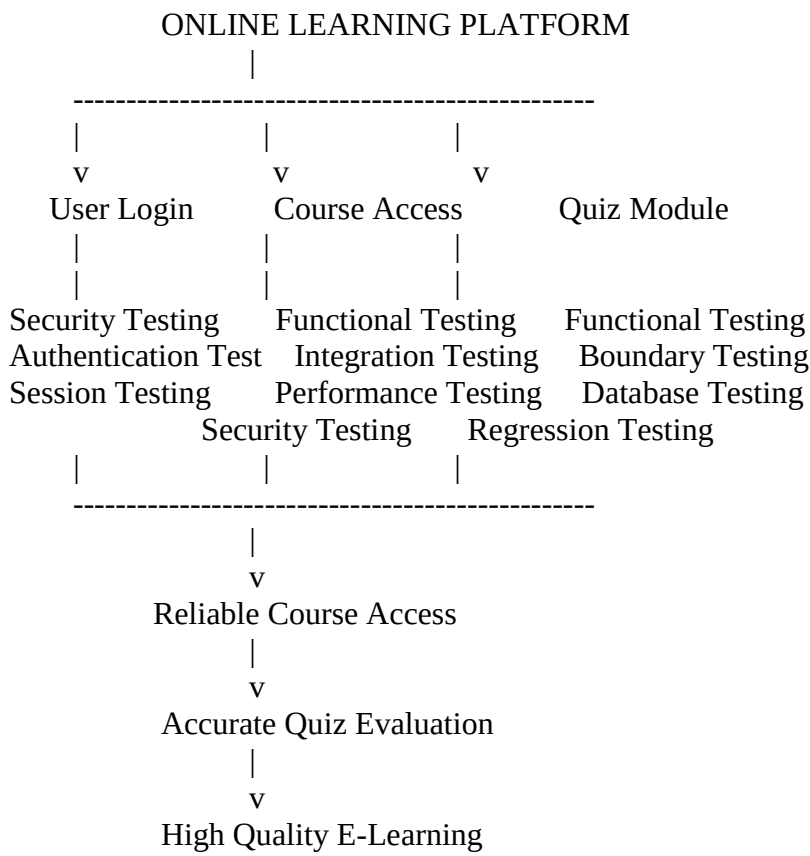
QA Verification

- Verify database records.
- Confirm result retrieval.
- Check report generation.

QA Techniques Summary Table

QA Technique	Purpose	Applied To
Functional Testing	Verify functionality	Course Access, Quiz Module
Integration Testing	Verify module interaction	Login & Course Access
Performance Testing	Measure speed and response	Course Videos
Security Testing	Protect data and resources	Login & Course Access
Boundary Value Testing	Test extreme values	Quiz Scores
Database Testing	Verify data storage	Quiz Results
Regression Testing	Ensure updates do not break features	Entire System
User Acceptance Testing	Validate user requirements	Course Access & Quiz Module

Quality Assurance Process Diagram



Conclusion

Quality Assurance techniques are essential for ensuring the reliability and effectiveness of an Online Learning Platform. Functional, Integration, Performance, and Security Testing help guarantee smooth and secure course access, while Functional, Boundary Value, Database, Regression, and User Acceptance Testing ensure accurate quiz evaluation. Through systematic testing and continuous quality monitoring, the platform can provide reliable learning services, correct assessment results, enhanced user satisfaction, and improved overall software quality.